

Integer Occurrences in Rational Linear Recurrences

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The Big Question

Integer Coefficients

$$S_0 = 1, S_1 = 7$$

$$S_{n+2} = 3S_{n+1} + 2S_n$$

Clearly all terms are integers.

Rational Coefficients?

$$S_0 = 1, S_1 = 1$$

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n$$

Can we get integers?

When do integers appear in rational recurrences?

Our Setting

Rational Second-Order Linear Recurrence

Given $S_0, S_1 \in \mathbb{Z}$, define

$$S_{n+2} = c_1 S_{n+1} + c_0 S_n$$

where $c_1 = \frac{c}{d}$ and $c_0 = \frac{a}{b}$ are **irreducible** fractions.

We study: $b > 1$ or $d > 1$ (at least one rational coefficient)

When
integers
appear?

Consecutive
integers?

Isolated
integers?

Finitely
many?

Example 1: Simple Case

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n \text{ with } S_0 = 2, S_1 = 3$$

n	S_n	Integer?
0	2	YES
1	3	YES
2	2	YES
3	$\frac{13}{6}$	NO
4	$\frac{31}{18}$	NO
5	$\frac{121}{108}$	NO

Three consecutive integers! Then no more.

The Key Transformation

Main Idea: Convert to an integer sequence to study divisibility.

Let $Q = \text{lcm}(b, d)$. Define the **auxiliary sequence**:

$$T_n := Q^n S_n$$

Key Properties

- (T_n) satisfies an **integer** recurrence: $T_{n+2} = PT_{n+1} + RT_n$
- Where $P = c \cdot \frac{Q}{d}$ and $R = a \cdot Q \cdot \frac{Q}{b}$ are both integers
- **Critical:** $S_n \in \mathbb{Z} \Leftrightarrow Q^n \mid T_n$

Integrality problem $\rightarrow$ Divisibility problem

Example 2: Auxiliary Sequence in Action

For $S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n$: $Q = 6$, $P = 2$, $R = 18$

Auxiliary recurrence: $T_{n+2} = 2T_{n+1} + 18T_n$ with $T_0 = 2$, $T_1 = 18$

n	T_n	$Q^n = 6^n$	$Q^n \mid T_n?$	$S_n = T_n/Q^n$
0	2	1	YES	2
1	18	6	YES	3
2	72	36	YES	2
3	468	216	NO	$\frac{13}{6}$
4	4104	1296	NO	$\frac{57}{18}$

Check: $72 = 36 \cdot 2$ $468 = 216 \cdot \frac{13}{6}$

p -adic Valuations

Tool: Use p -adic valuations $\nu_p(x) =$ highest power of p dividing x

p -adic Integrality Criterion

$$S_n \in \mathbb{Z} \Leftrightarrow \nu_p(T_n) \geq n\nu_p(Q) \text{ for every prime } p \mid Q$$

Example: $Q = 6 = 2 \cdot 3$

Need **both**:

- $\nu_2(T_n) \geq n$ (enough factors of 2)
- $\nu_3(T_n) \geq n$ (enough factors of 3)

This allows us to analyze each prime separately!

Connection to Lucas Sequences

The auxiliary sequence connects to **Lucas sequences**:

Lucas Sequence

$$\begin{cases} U_0 = 0 \\ U_1 = 1 \\ U_{n+2} = PU_{n+1} + RU_n \end{cases}$$

Companion Identity

$$T_n = T_1 U_n + R T_0 U_{n-1}$$

Why this is powerful:

- Lucas sequences have well-studied p -adic properties
- Results from Ballot [1] give exact formulas for $\nu_p(U_n)$
- We can use these to understand when T_n has enough divisibility

Finding Long Runs of Integers

Theorem (Constructive Initial Integrality)

For any $m \geq 1$ and any rational coefficients, we can find $S_0, S_1 \in \mathbb{Z}$ such that

$S_0, S_1, S_2, \dots, S_m$ are **all** integers

How? Choose initial conditions with **very high** p -adic valuations:

$$S_0 = \prod_{p|Q} p^{A_{0,p}(m)}, \quad S_1 = \frac{1}{Q} \prod_{p|Q} p^{A_{1,p}(m)}$$

where the exponents are carefully computed from Lucas sequence valuations.

Key insight: Start with enough factors of each prime!

Example 3: Computational Results

$$S_{n+2} = \frac{23}{2}S_{n+1} + 70S_n \text{ with carefully chosen } S_0, S_1$$

Powers of 2 dividing S_n (i.e., $\nu_2(S_n)$) for different initial conditions:

Initial conditions	ν_2 pattern
(81, 76)	0, 2, 4, 6, 8, 10, 12, 15 , 13, 12, ..., 0 22 terms!
(6247, -7084)	0, 2, 4, 6, 8, 10, 12, 14, 16, 16 , 15, ..., 0 26 terms!
(20037, -20036)	0, 2, 4, 6, 8, ..., 18, 20 , 20, 19, ..., 0 32 terms!
(301487, -247884)	0, 2, 4, 6, 8, ..., 20, 22 , 22, 21, ..., 0 35 terms!

Pattern increases with larger initial conditions!

Visualizing the Pattern

$$S_{n+2} = \frac{23}{2}S_{n+1} + 70S_n \text{ with } (81, 76)$$

Binary representation of terms:

$$S_0 = 81 = 1010001_2$$

$$S_1 = 76 = 1001100_2$$

$$S_2 = 6544 = 1100110010000_2$$

$$S_3 = 80576 = 10011101011000000_2$$

$$S_4 = 1384704 = 101010010000100000000_2$$

$\vdots$

Notice the trailing zeros increase (these are the powers of 2)!

The pattern: grows, peaks at $n = 7$, then decreases back to 0 at $n = 21$.

Combining Solutions: The Gluing Method

Can we extend integer runs even further?

Idea: If we have two sequences with runs of length m_0 and m_1 , combine:

$$\text{New sequence} = u \cdot (\text{Sequence 0}) + v \cdot (\text{Sequence 1})$$

for carefully chosen integers u, v .

Example

Starting from $(11, 4)$ with 11 integer terms and $(5, -4)$ with 8 terms:

$$8 \cdot (11, 4) + (11, 4) = (51, -28)$$

gives a sequence with **13 integer terms!**

This allows iterative construction of longer and longer runs.

Restrictions on Consecutive Integers

Not everything is possible!

Theorem (No Consecutive Integers)

If $\gcd(b, d) = 1$, $b > 1$, $d > 1$, and $\gcd(S_0, S_1) = 1$, then:

S_n and S_{n+1} **cannot both** be integers for any $n \geq 2$

Intuition: The coprimality forces divisibility constraints that propagate backward. If two consecutive terms far out are integers, we can show S_0 and S_1 must share a common factor, contradicting $\gcd(S_0, S_1) = 1$.

Example 4: No Consecutive Integers

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n \text{ with } S_0 = 1, S_1 = 1$$

Here $b = 2, d = 3$, so $\gcd(b, d) = 1$ and $\gcd(1, 1) = 1$

n	S_n	Status
0	1	Integer
1	1	Integer
2	$\frac{5}{6}$	Not integer
3	$\frac{7}{9}$	Not integer
4	$\frac{52}{81}$	Not integer

Confirmed: No consecutive integers for $n \geq 2$

Definition

A prime p is **special** for (P, R) if both $p \mid P$ and $p \mid R$.

When p is special, $\nu_p(U_n)$ has predictable structure (from Ballot [1]):

Let $a = \nu_p(P)$, $b = \nu_p(R)$. If $b < 2a$:

$$\nu_p(U_{2n+1}) = bn$$

$$\nu_p(U_{2n}) = bn + (a - b) + \nu_p(n) + \lambda_n$$

Key observation: The valuation at **even** indices grows faster!

This creates opportunities for isolated integers at specific positions.

Theorem (Prescribed Integers)

Under suitable conditions on the primes, we can construct S_0, S_1 such that

$$S_{2m_1}, S_{2m_2}, \dots, S_{2m_k} \text{ are integers}$$

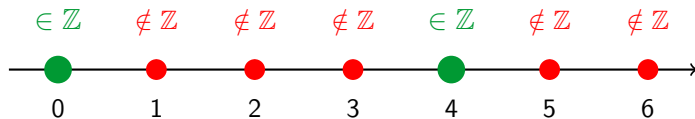
for **any** specified even indices $2m_1 < 2m_2 < \dots < 2m_k$!

Method:

- 1 Each index m_j gives a congruence condition on $S_0/S_1 \pmod{p^{L(m_j)}}$
- 2 Space indices appropriately so congruences are compatible
- 3 Use Chinese Remainder Theorem to satisfy all conditions simultaneously

Example 5: Integer Islands

Conceptual example: Create integers only at $n = 0$ and $n = 4$



Even with $\gcd(S_0, S_1) = 1$, we can create isolated integers at prescribed locations using p -adic congruences!

Only Finitely Many Integers

For a **fixed** sequence, integers can't keep appearing forever.

Theorem (Finiteness)

Fix the recurrence and initial conditions S_0, S_1 . If the characteristic roots satisfy r_1/r_2 not a root of unity, then there exists explicit N such that

$$S_n \notin \mathbb{Z} \text{ for all } n > N$$

Why? Using Binet formula $S_n = \alpha r_1^n + \beta r_2^n$:

- Integrality requires $\nu_p(\Gamma \Lambda^n + 1) \geq np - C$ (linear growth in n)
- But p -adic linear forms theorem (Chim [2]) gives upper bound $\leq K(\log n + C')$ (logarithmic!)
- Linear eventually beats logarithmic $\Rightarrow$ explicit cutoff

Example 6: Explicit Cutoff

$$S_{n+2} = \frac{5}{2}S_{n+1} + \frac{3}{4}S_n \text{ with } S_0 = 1, S_1 = 1$$

Characteristic roots: $r_1 = 3$, $r_2 = -\frac{1}{4}$

Since $Q = 4 = 2^2$ and $\nu_2(r_2) = -2$, we need $\nu_2(S_n) \geq 0$, which forces

$$\nu_2(\Gamma\Lambda^n + 1) \geq 2n - \text{constant}$$

But the p -adic logarithm bound gives:

$$\nu_2(\Gamma\Lambda^n + 1) \leq K_2(\log n + C_H)$$

Solving $2n - C_1 \leq K_2(\log n + C_H)$ gives **explicit** N .

Conclusion: Only finitely many integers in this sequence!

What We Proved About Integer Occurrences

Construction: Can create arbitrarily long runs of consecutive integers at the start (choose initial conditions with high p -adic valuations)

Restriction: With coprime denominators and $\gcd(S_0, S_1) = 1$, no two consecutive terms beyond S_2 can both be integers

Islands: Under special prime conditions, can prescribe isolated integers at arbitrary even indices

Finiteness: For any fixed sequence, only finitely many integers appear (with explicit computable bound)

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Thank you!

Questions?